

PKCERT AI & Software Development Internship

Task 10 – Cross-Validation & Hyperparameter Tuning

Objective:

Develop practical skills in improving machine learning models using Cross-Validation and Hyperparameter Tuning techniques, including GridSearchCV and RandomizedSearchCV.

Total Marks: 100

Part A – Dataset Preparation (15 Marks)

1. Select a public classification dataset.
2. Describe the dataset, features, and target variable.
3. Preprocess the dataset and split it into training and testing sets.

Part B – Cross-Validation (30 Marks)

1. Train a classification model.
2. Apply K-Fold Cross-Validation.
3. Interpret the cross-validation results.

Part C – Hyperparameter Tuning (40 Marks)

1. Optimize the model using GridSearchCV.
2. Optimize the same model using RandomizedSearchCV.
3. Compare the optimal parameters and evaluate using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Part D – Comparative Analysis (15 Marks)

1. Compare the baseline and optimized models.
2. Discuss the advantages and limitations of GridSearchCV and RandomizedSearchCV.
3. Recommend the most suitable approach with justification.

Assessment Area	Marks
Dataset Preparation	15
Cross-Validation	30
Hyperparameter Tuning	40

